BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Standards and Protocols

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What is SMTP used for?

2. Unsigned 8-bit values can reach 255. Does 248 + 16 overflow?

A) Yes: the sum is 264
B) No: the sum is 264
C) No: the sum is 8
D) Yes: the sum is 232

3. What is the Internet?

4. A browser asks for a school's web page. "A client must host a website before asking for it." Correct this. Which computer provides the page?

Today's objectives

Explain why agreed network standards are useful. Distinguish standards from communication protocols.

Key vocabulary - Read, cover, recall

Standard
Agreed technical specification allowing compatibility.

Protocol
Agreed rules for communication between devices.

Interoperability
Different hardware/software systems working together.

HTTP
Protocol for web communication.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Why standards matter	Standards are agreed rules/specifications that allow equipment and software from different manufacturers to communicate.
Protocols in action	A protocol provides rules for exchanging data. Different services need different protocol functions.
Different communication jobs	A web request uses a web protocol such as HTTP/HTTPS. TCP supports reliable ordered delivery, while IP handles addressing and routing.
Layering is an explanation	Communication tasks can be considered in stages/layers. OCR J277 does not require memorising names and functions of all TCP/IP model layers; focus on the required standards and protocol functions.

I Do - One worked example

A school purchases equipment from several manufacturers. Explain why standards and communication protocols help it work together. [4 marks]

1. Different manufacturers need compatible rules if their devices are to communicate.

2. Agreed standards [1] help equipment from different makers interoperate [1].

3. Communication protocols define common rules for sending data [1], so requests can be interpreted correctly by devices [1].

4. Four points: standard -> compatibility, protocol -> effective exchange.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A tablet from one company uses a website hosted by another. Explain why communication standards matter. [4 marks]

2. A client requests a web page. Explain the separate roles of HTTP and TCP. [4 marks]

3. Explain how TCP and IP work together when sending data between networks. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Write the second print instruction.

```
print("Request")
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. State one benefit of agreed networking standards. [1 mark]

Q3. A public-transport ticket kiosk communicates with a booking service made by a different manufacturer. Explain two benefits of both systems following agreed communication standards or protocols. [4 marks]

Q4. Explain how standards and HTTP, TCP and IP help successful network communication. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. OCR J277 requires memorising all four TCP/IP layer names and their functions. ☐ **T** ☐ **F**
2. Standards are agreed rules/specifications that allow equipment and software from different manufacturers... ☐ **T** ☐ **F**
3. A protocol provides rules for exchanging data ☐ **T** ☐ **F**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Sending email from a client to a mail server and between mail servers. ☐

Q2 A - Yes: the sum is 264 ☐

Q3 A worldwide collection of interconnected computer networks. ☐

Q4 A web server stores website resources and sends them when clients request them. Access needs working servers and network services. ☐

Code First - Model

```
print("Request")
print("Response")
```

Output: Request
Response

We Do - Check the model

1. Both devices follow agreed networking rules [1] even though built by different manufacturers [1], so they can exchange compatible data [1] and use network services [1]. ☐

2. HTTP sets rules for requesting/transferring web content [1] so client/server know which resource is needed [1]. TCP helps deliver data reliably [1] in correct order [1]. ☐

3. TCP supports reliable delivery [1] and correct ordering [1]; IP uses destination network addressing [1] and routing between networks [1]. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 They allow equipment/software from different manufacturers to communicate/interoperate [1]. ☐

Q3 Both systems use the same agreed message rules [1], so requests/responses are understood despite different manufacturers [1]. Shared standards enable interoperability [1], avoiding a separate custom communication system for each manufacturer's equipment [1]. ☐

Q4 Standards support device compatibility [1] between different manufacturers [1]. HTTP defines web requests and responses [1]. TCP supports reliable arrival [1] and correct order [1]. IP supplies destination addressing/routing [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: